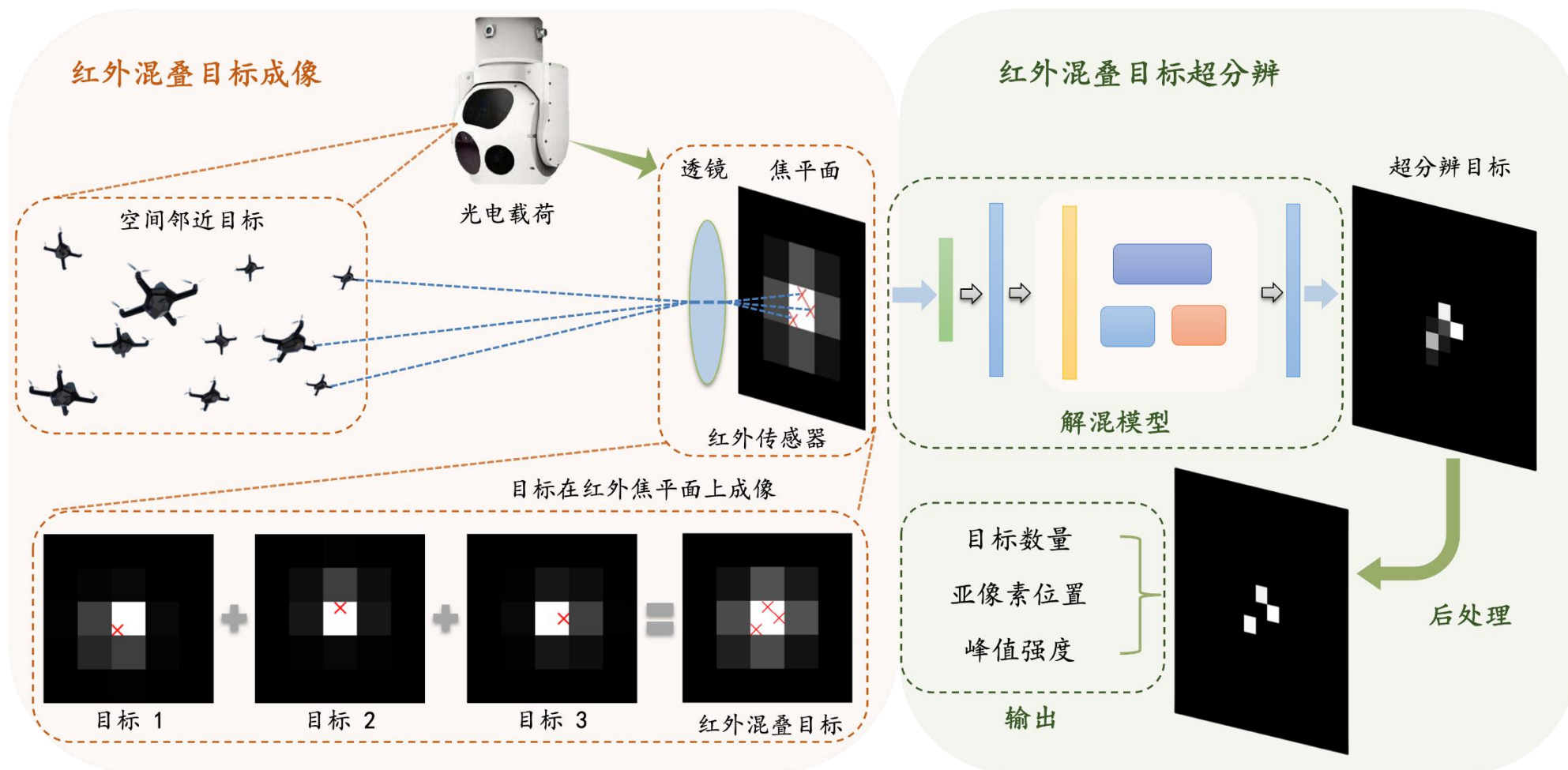


空间邻近目标解混

稀疏 $\rightarrow$ 稠密 $\rightarrow$ 混叠



空间邻近的红外弱小目标 (ICCV 2025、TGRS 2025)

空间邻近目标解混

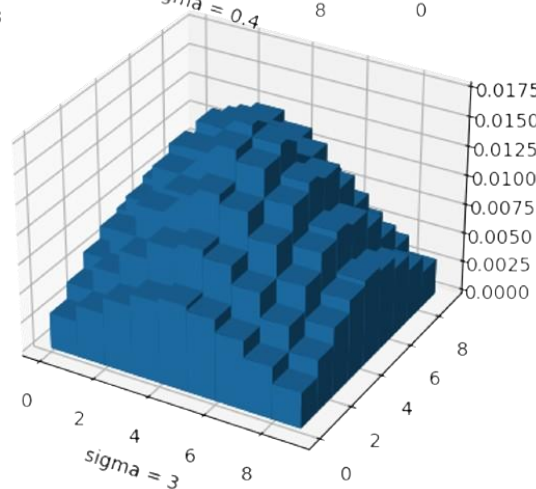
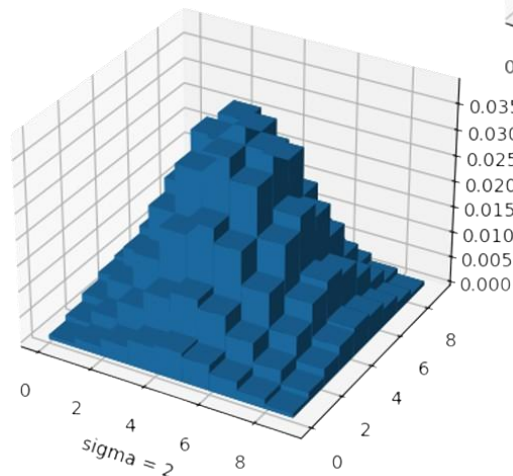
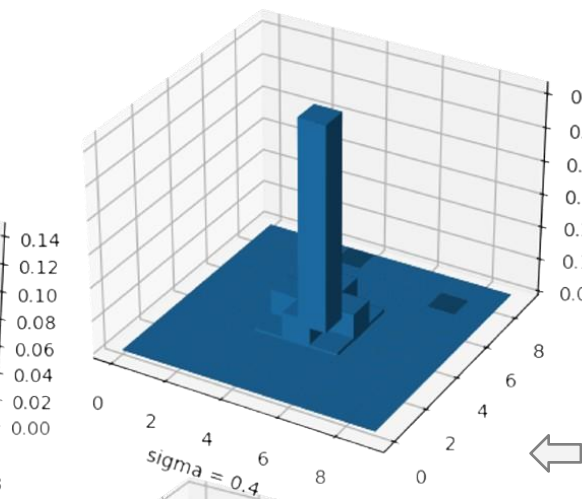
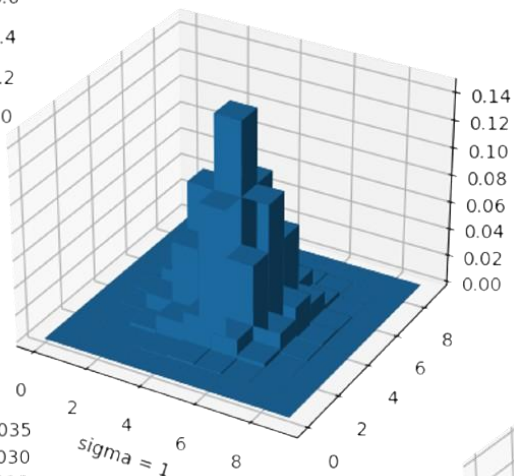
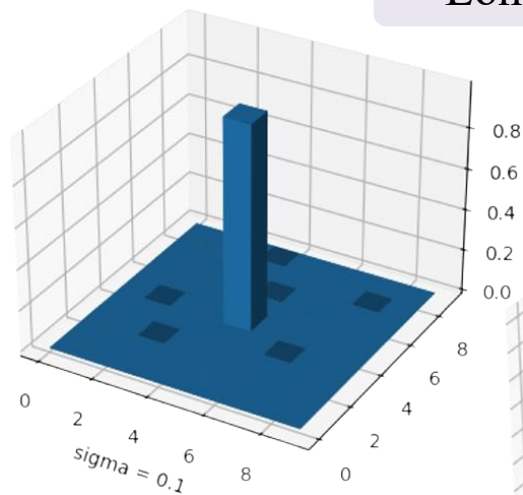
Long-Distance



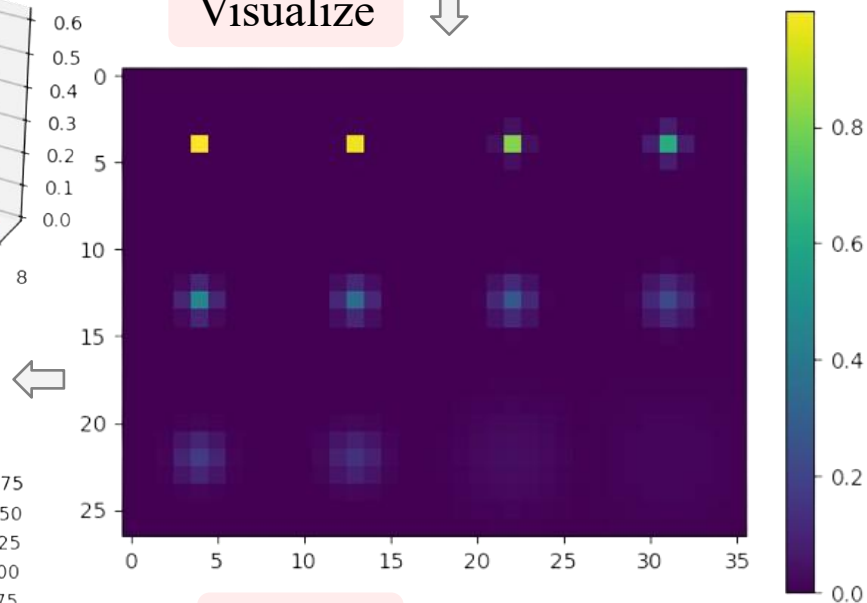
Point Light Source



高斯函数表示点目标



Visualize



Imaging

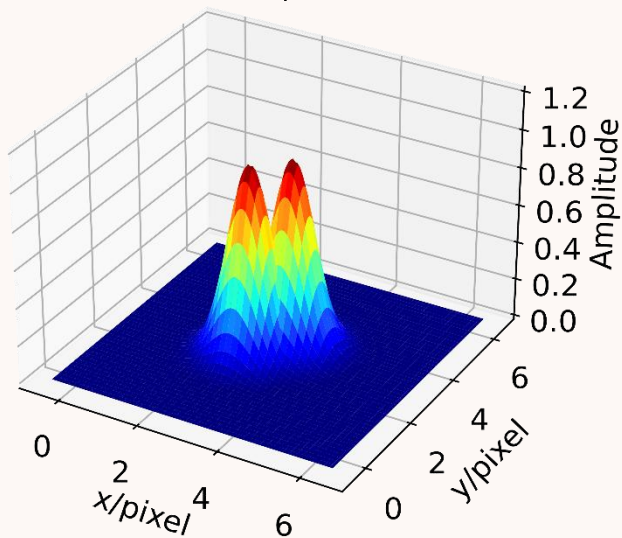
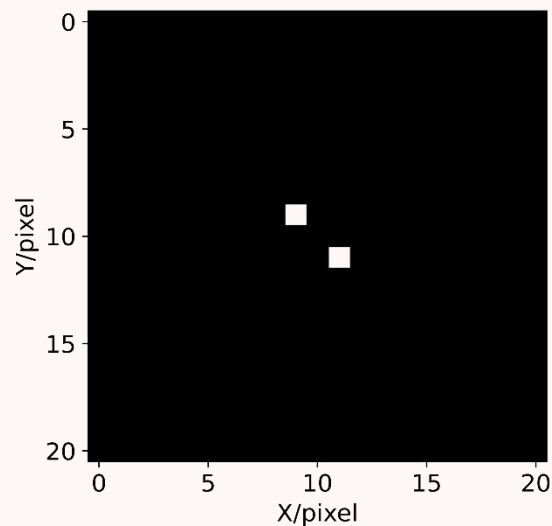


$$g_{i,j}(x_t, y_t) = \int_{x_{i,j}-\frac{1}{2}D}^{x_{i,j}+\frac{1}{2}D} \int_{y_{i,j}-\frac{1}{2}D}^{y_{i,j}+\frac{1}{2}D} p(x, y) dx dy$$

像平面上进行能量积分

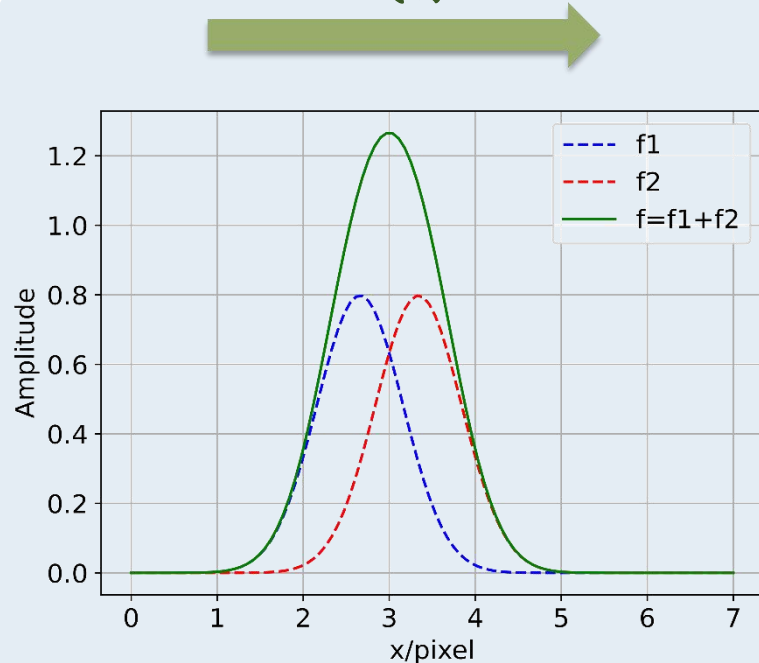
空间邻近目标解混

响应尖峰



独立 PSF 函数

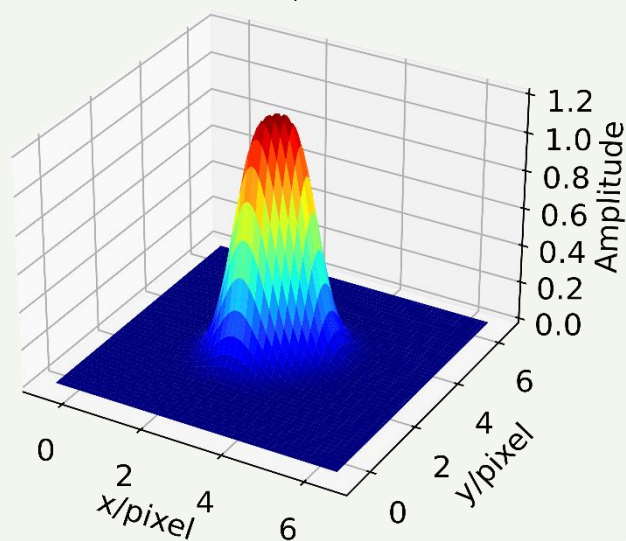
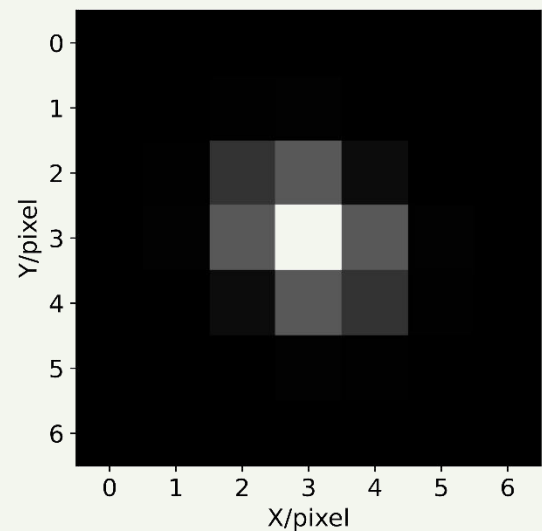
成像



线性混叠

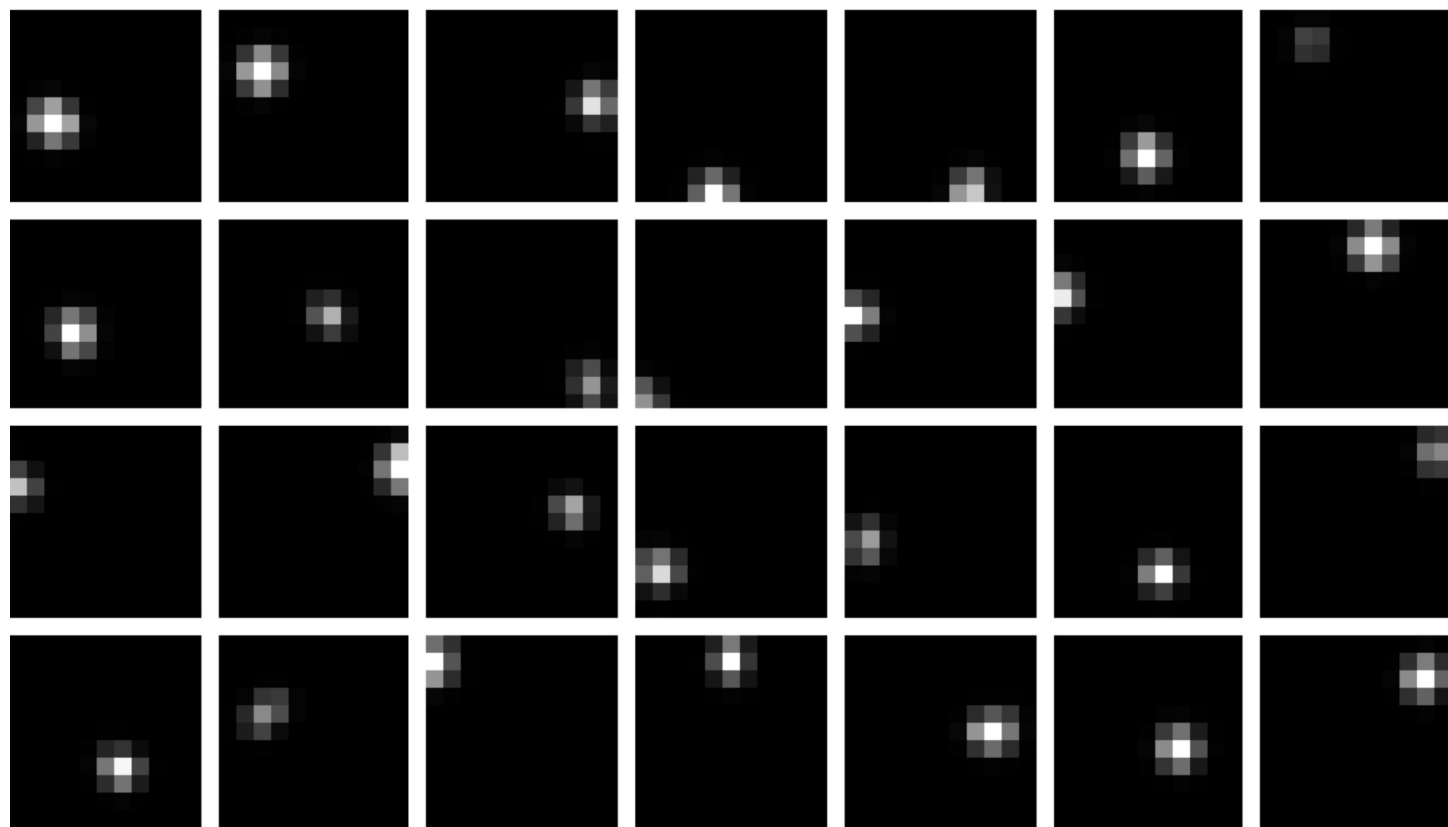
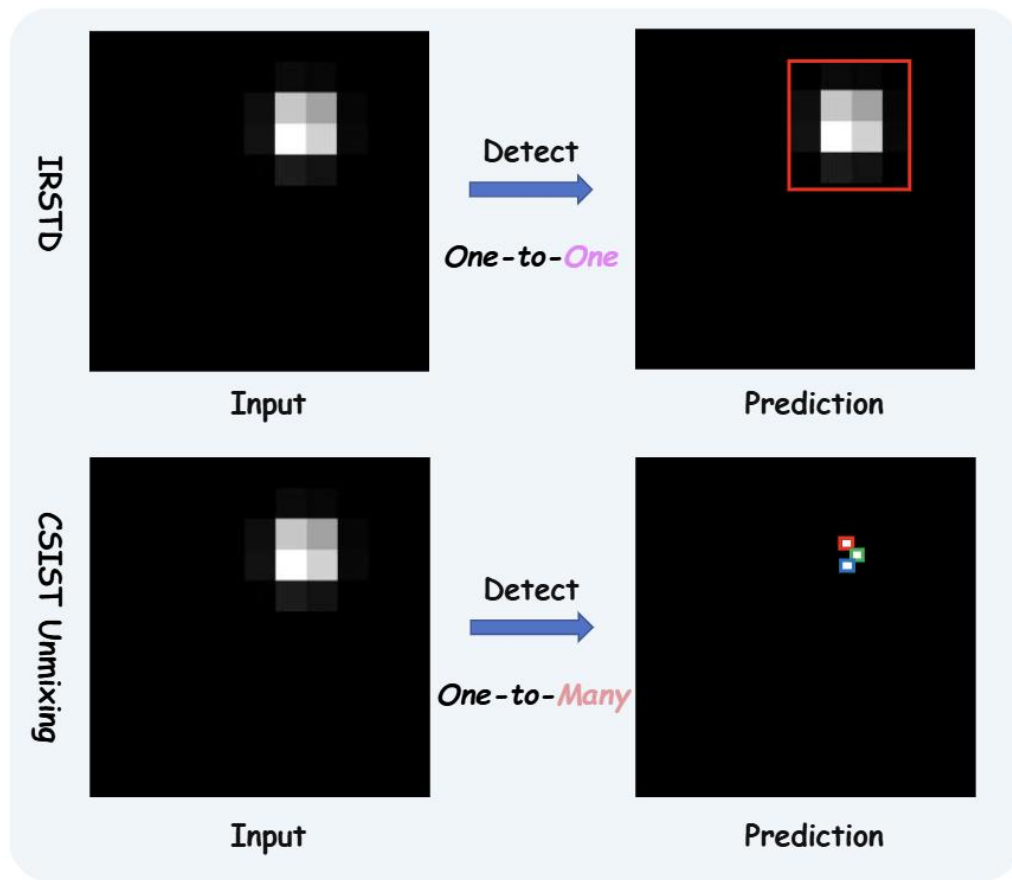
超分辨

混叠响应



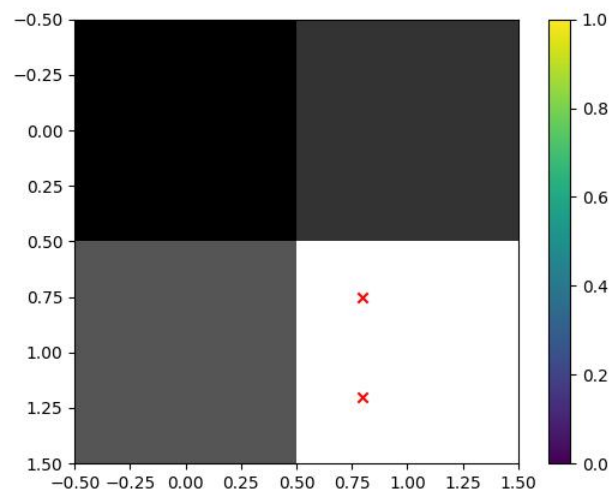
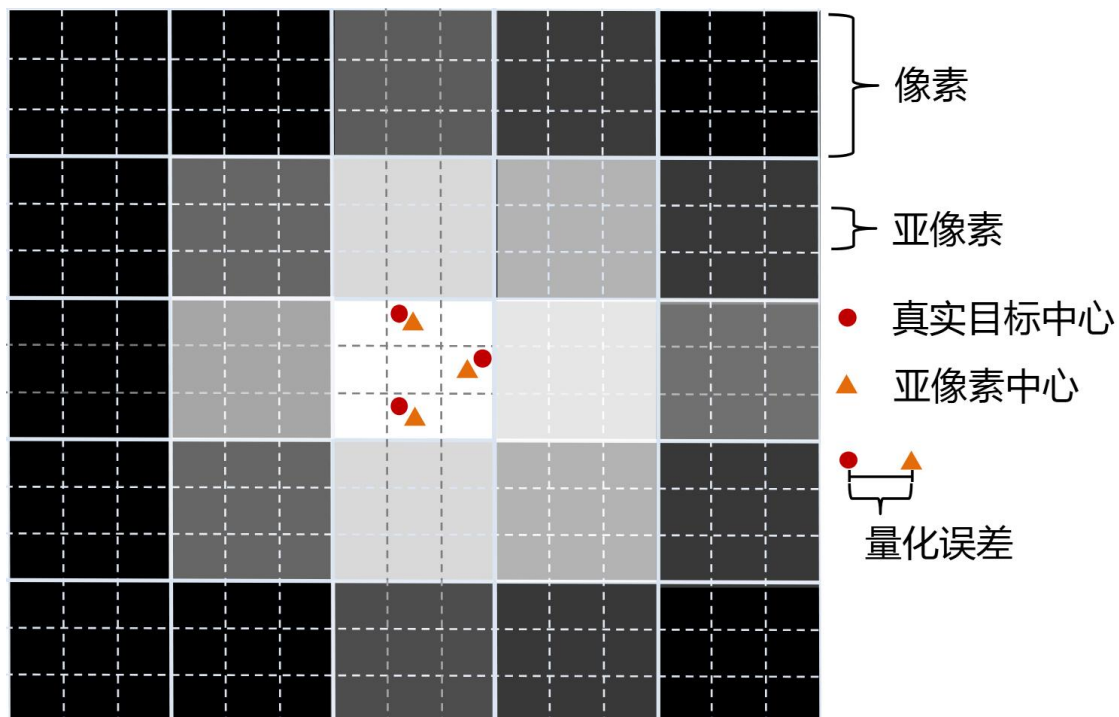
PSF 函数

空间临近目标解混



单帧 CSIST-100K Dataset

空间邻近目标解混



Sup-pixel Grid

$$g_{i,j}(x_t, y_t) = \int_{x_{i,j}-\frac{1}{2}D}^{x_{i,j}+\frac{1}{2}D} \int_{y_{i,j}-\frac{1}{2}D}^{y_{i,j}+\frac{1}{2}D} p(x, y) dx dy$$

$$\begin{pmatrix} z[0][0] \\ z[0][1] \\ z[1][0] \\ z[1][1] \end{pmatrix} = \begin{pmatrix} g_{(0,0)}(x1, y1) & \dots & g_{(0,0)}(x16, y16) \\ g_{(0,1)}(x1, y1) & \dots & g_{(0,1)}(x16, y16) \\ g_{(1,0)}(x1, y1) & \dots & g_{(1,0)}(x16, y16) \\ g_{(1,1)}(x1, y1) & \dots & g_{(1,1)}(x16, y16) \end{pmatrix} \times \begin{pmatrix} s(x1, y1) \\ \dots \\ s(x16, y16) \end{pmatrix}$$

Observation

Measurement

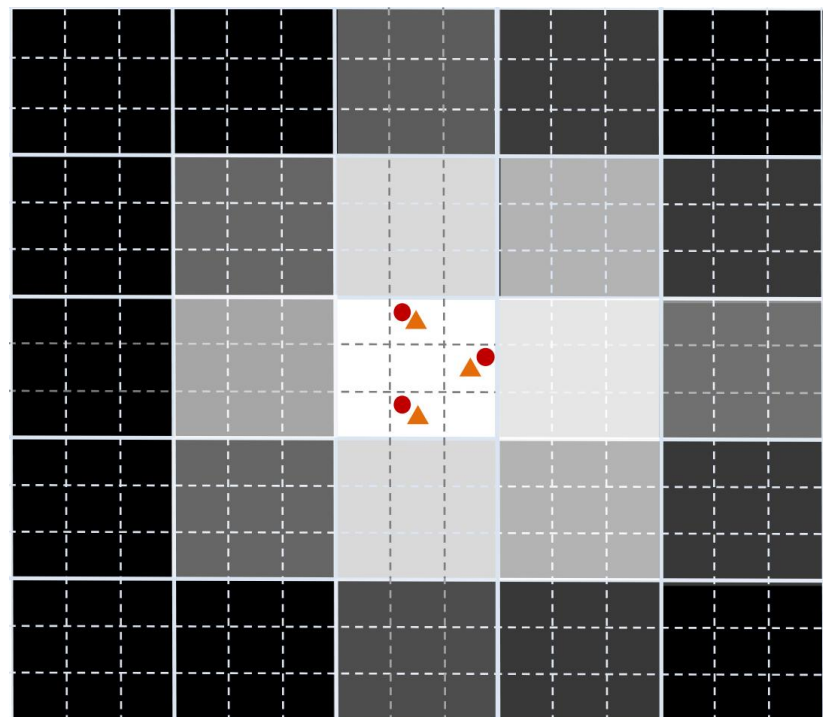
Intensity

Vectorize

$$\mathbf{z} = \mathbf{g}\mathbf{s} + \mathbf{n}$$

$$\arg \min_s \|\mathbf{y} - \mathbf{G}\mathbf{s}\|_2^2 + \lambda \|\mathbf{s}\|_1$$

空间邻近目标解混



像素

亚像素

● 真实目标中心

▲ 亚像素中心

量化误差

$$\min_{\tilde{\mathbf{s}}} \|\mathbf{z} - \mathbf{G}(\Omega)\tilde{\mathbf{s}}\|_2^2 + \lambda \|\tilde{\mathbf{s}}\|_1,$$

$$\mathbf{r}^{(k)} = \tilde{\mathbf{s}}^{(k-1)} - \rho \mathbf{G}^\top (\mathbf{G}\tilde{\mathbf{s}}^{(k-1)} - \mathbf{z})$$

$$\tilde{\mathbf{s}}^{(k)} = \arg \min_{\tilde{\mathbf{s}}} \frac{1}{2} \|\tilde{\mathbf{s}} - \mathbf{r}^{(k)}\|_2^2 + \lambda \|\Psi \tilde{\mathbf{s}}\|_1$$

Sub-pixel Division

稀疏表示优化求解
从凸优化到深度展开
GroCV
2024-01-15

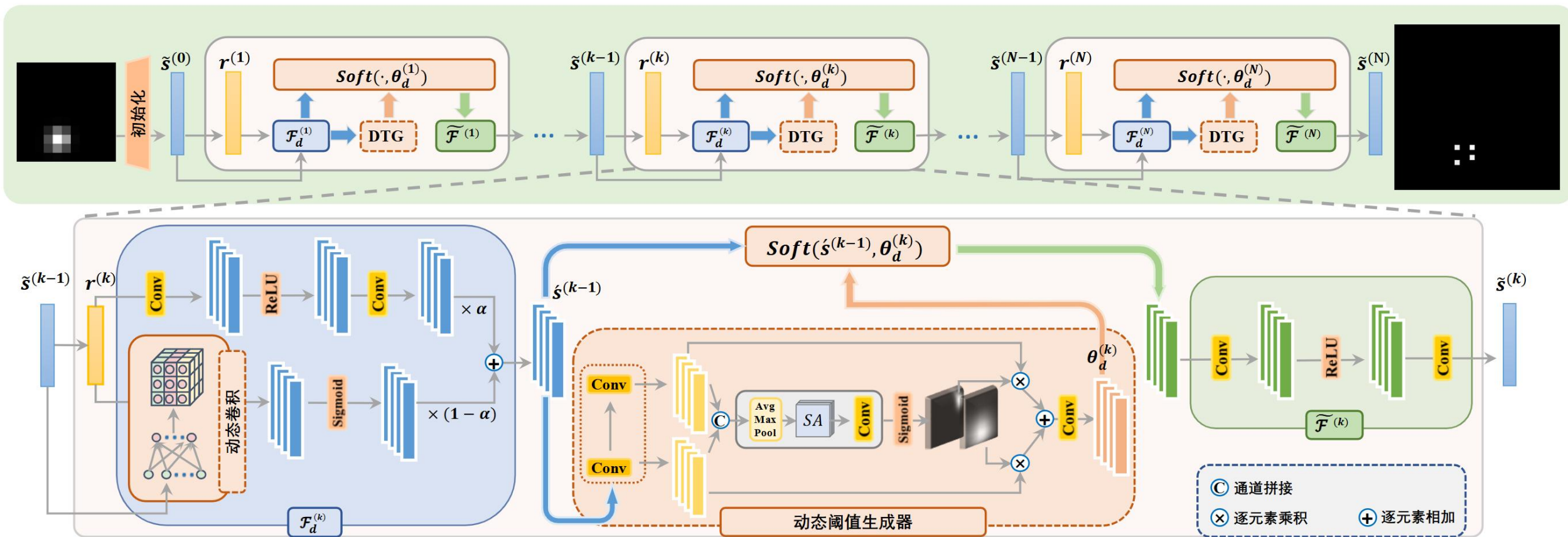
【文献梳理】稀疏表示优化求解：从凸优化到深度展开

2762 2 1-15

参考文献： [1] Daubechies I, Defrise M, De Mol C. An iterative problems with a sparsity constraint [J]. Commun. Pur. Appl. I

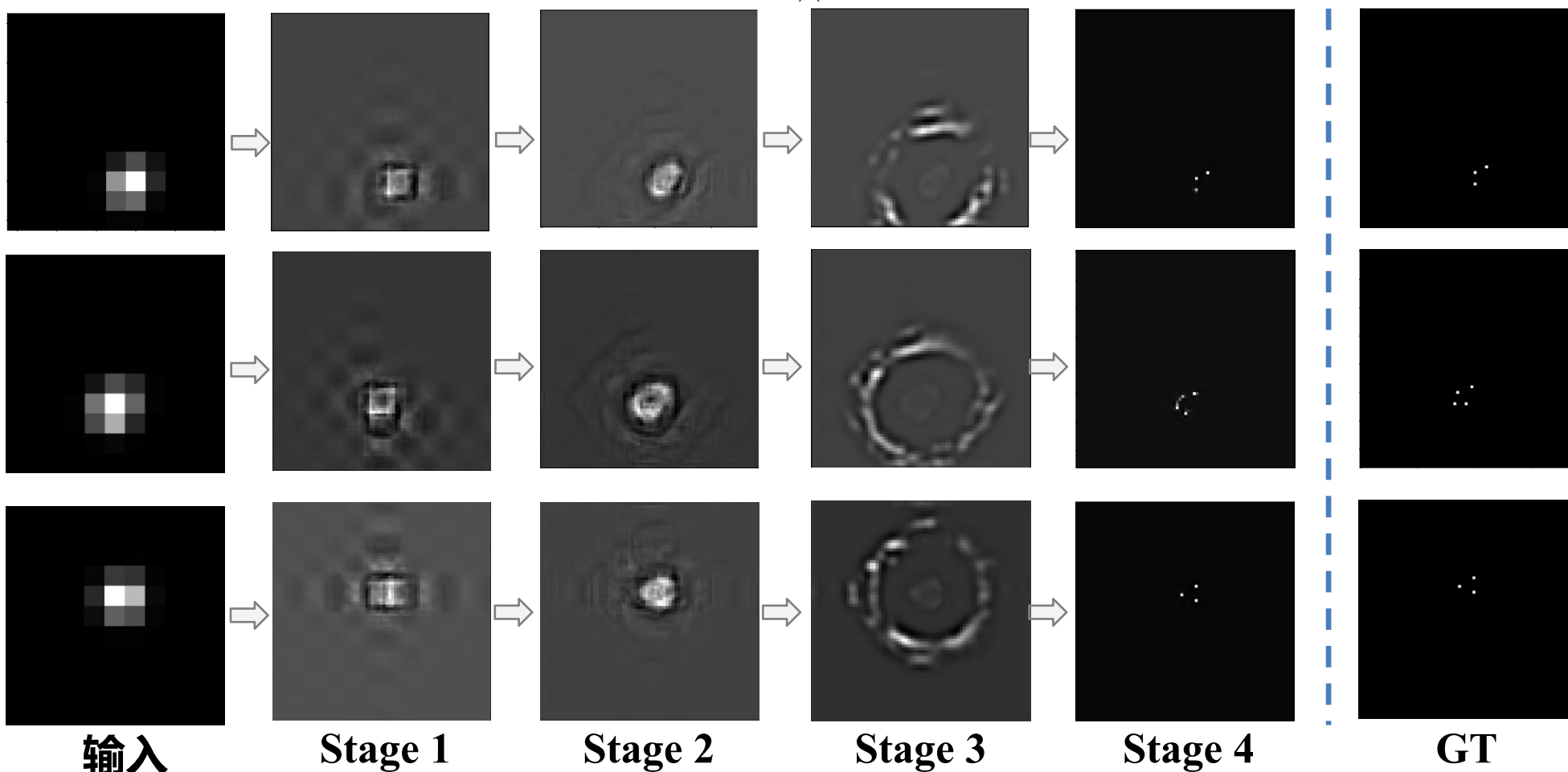
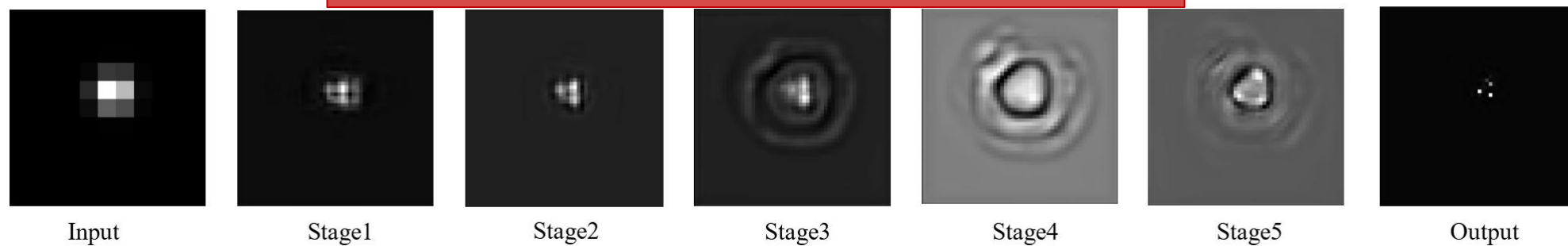
29:27

空间邻近目标解混

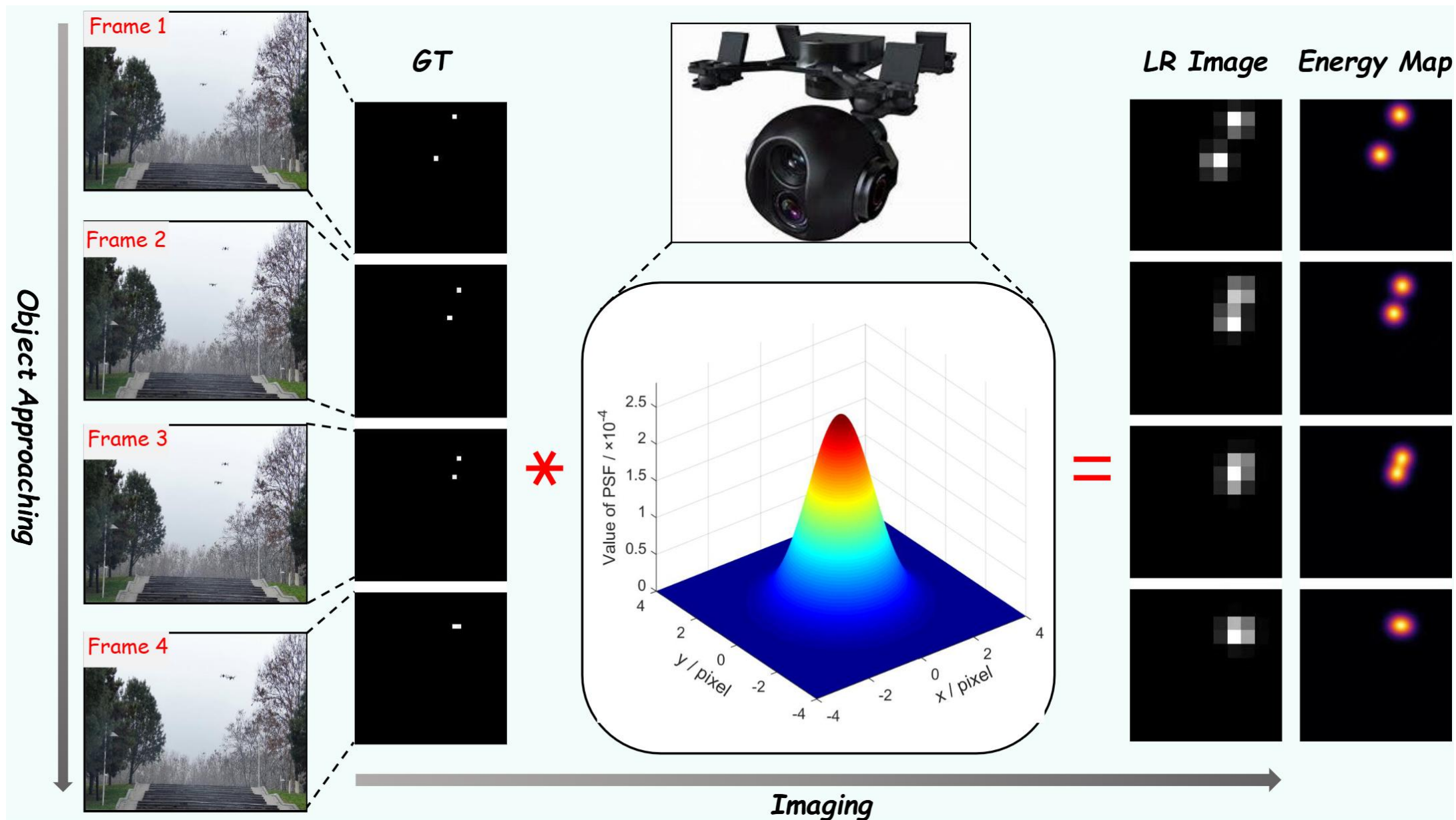


DISTA-Net

空间邻近目标解混

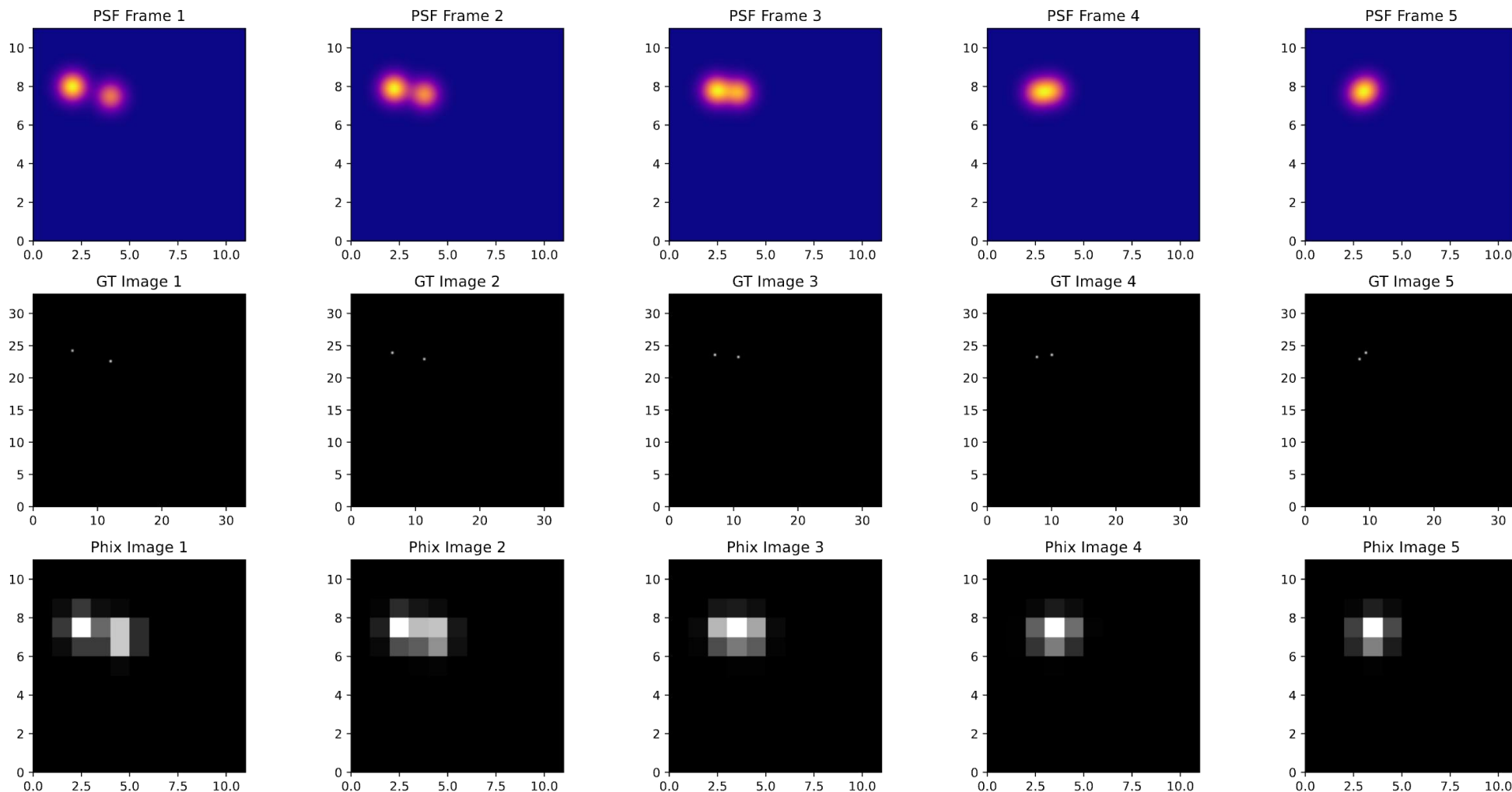


空间邻近目标解混



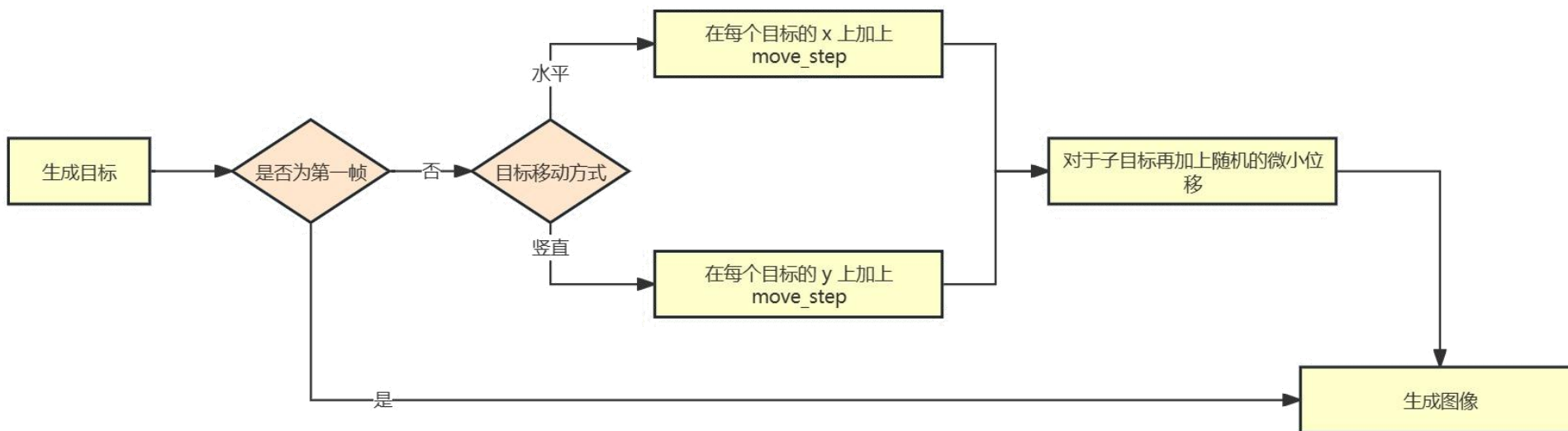
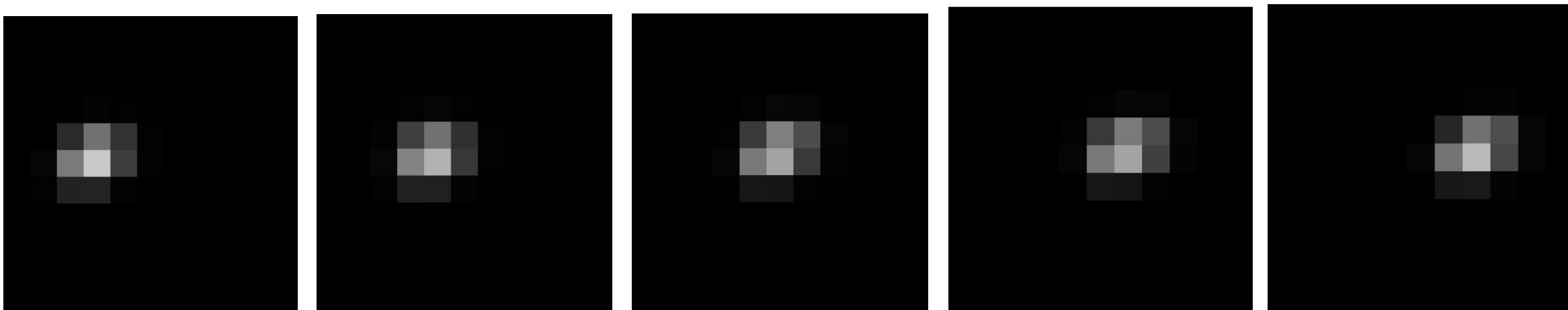
空间邻近目标解混

单帧  序列



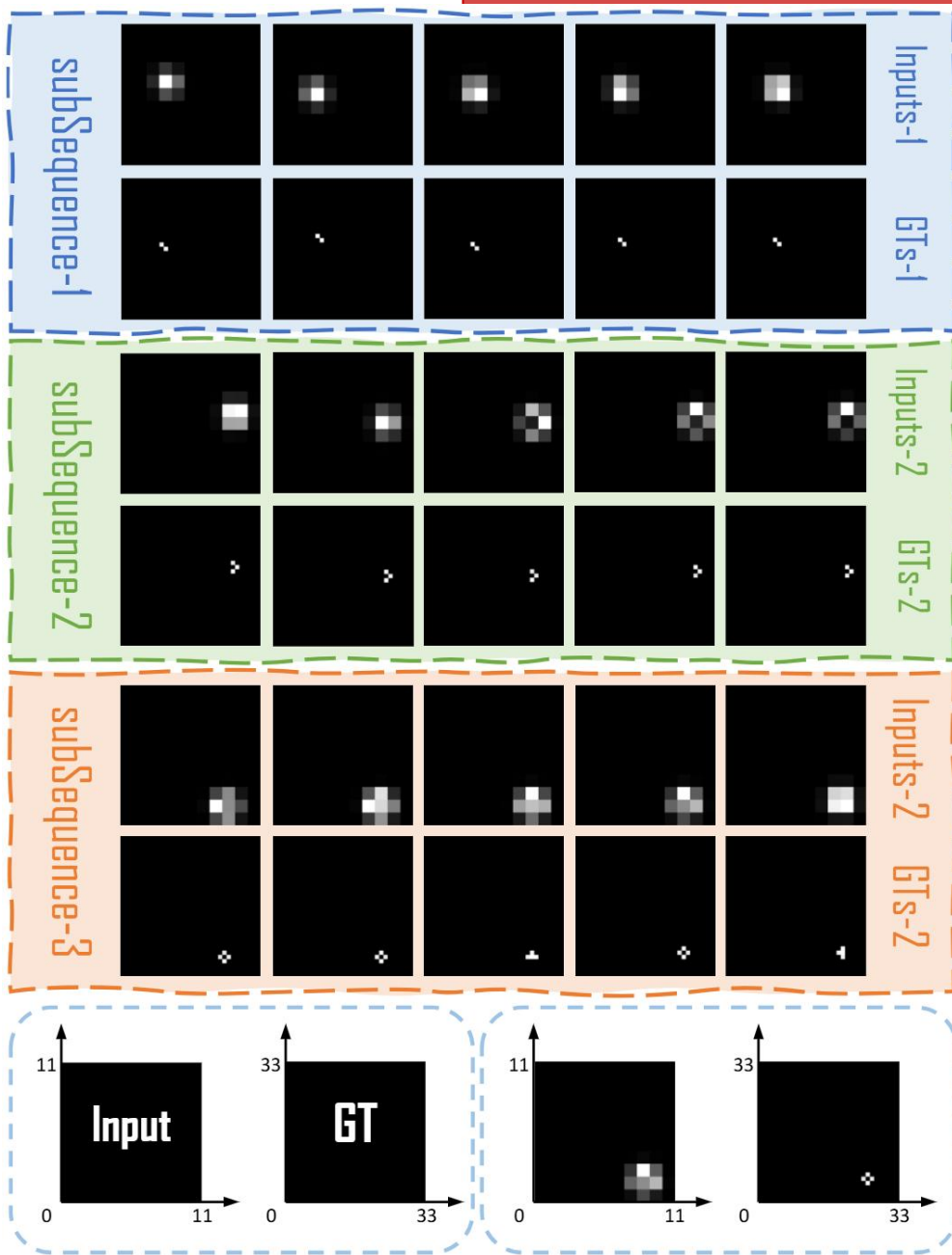
序列空间邻近目标解混

空间邻近目标解混



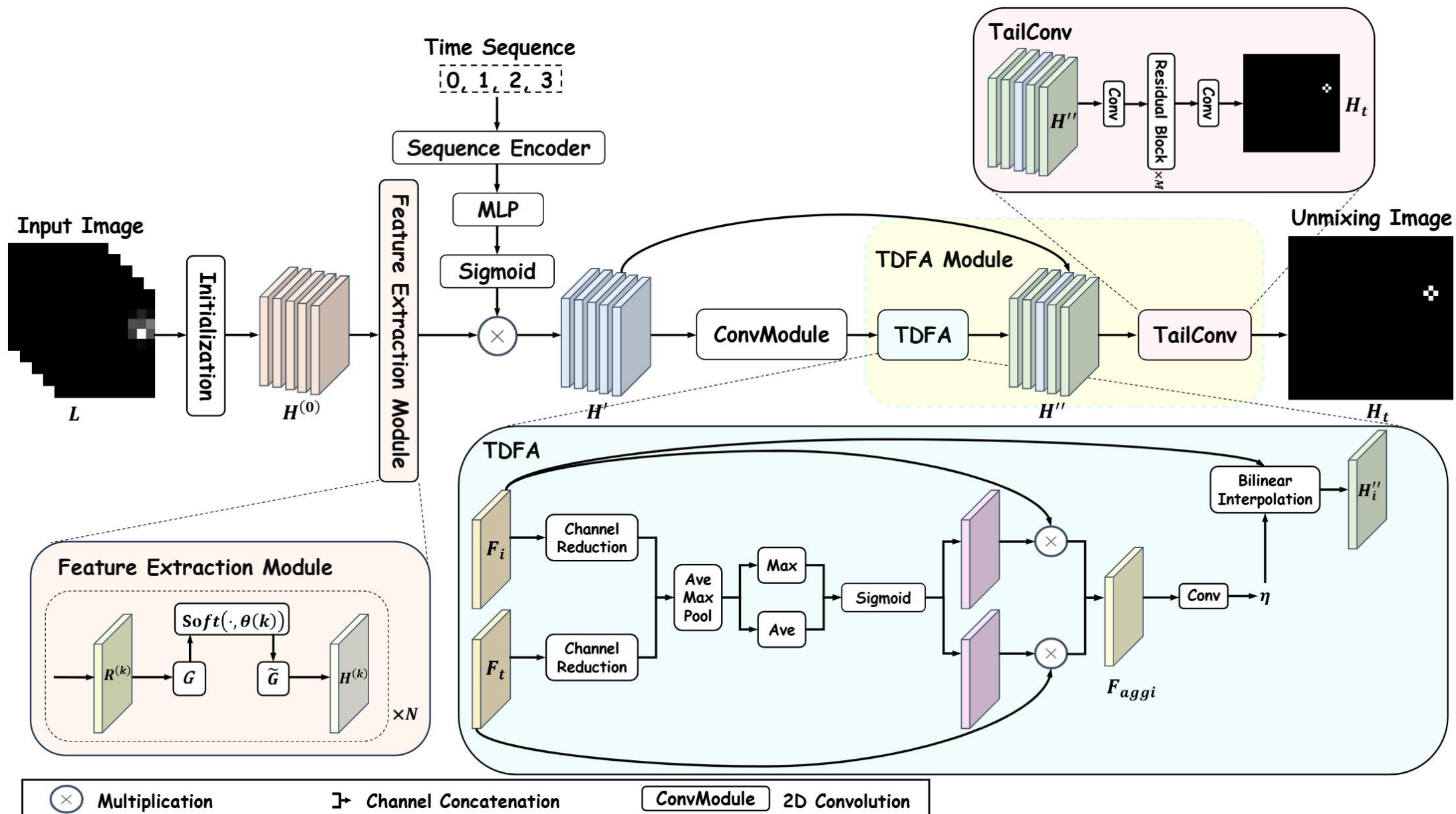
序列空间邻近目标生成

空间邻近目标解混

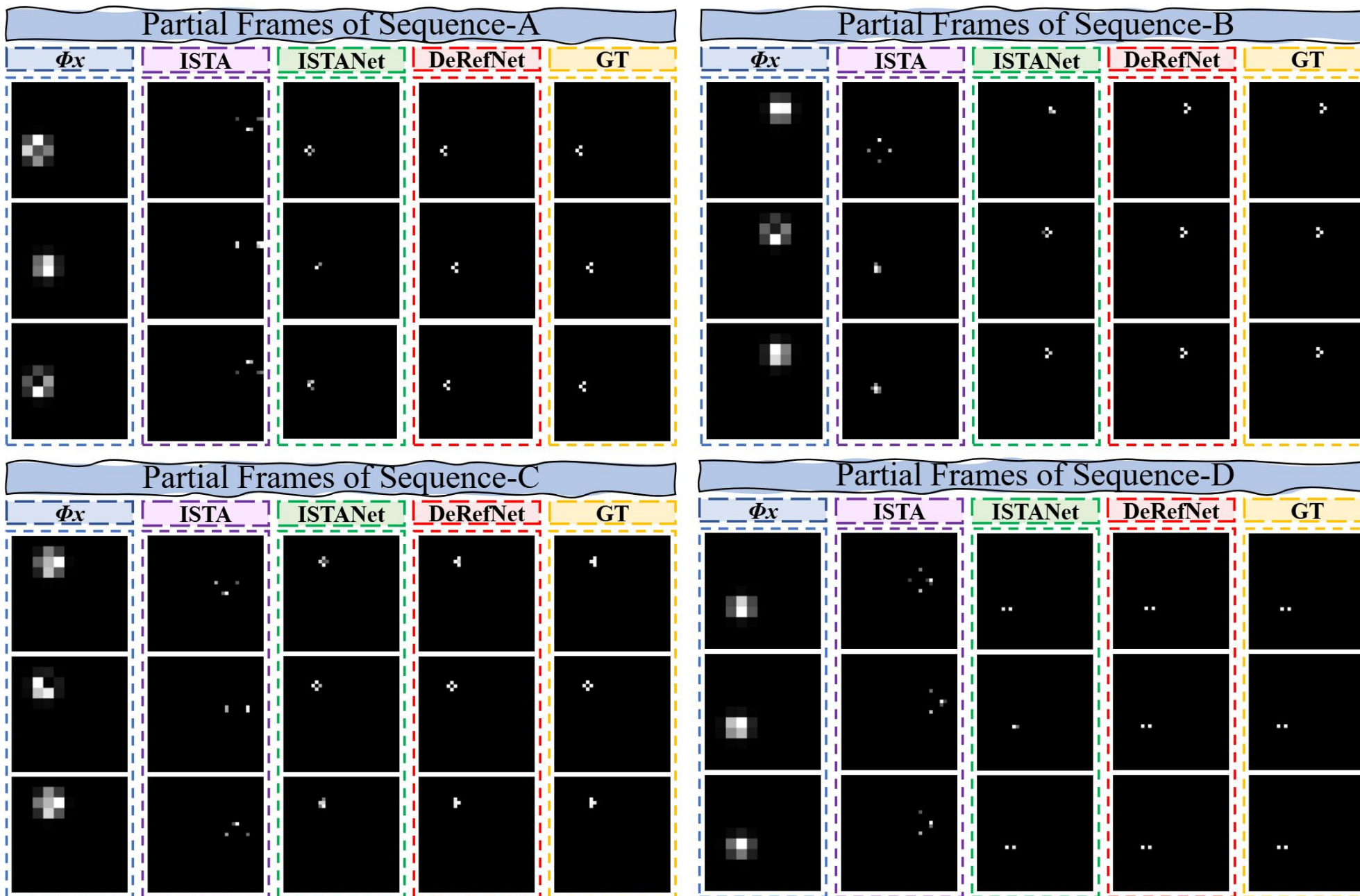


SeqCSIST 数据集

空间邻近目标解混



空间邻近目标解混



感谢聆听



GrokCV

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